

Functional Requirement Document

Based on the provided Business Requirement Document (BRD), I have created a detailed Functional Requirement Document (FRD) adhering to the "Coding Requirements" section. The requirements are written in bullet points.

FR-DP-001: Load Customer and Order Data from CSV to Delta Tables

Title: Load Customer and Order Data from CSV to Delta Tables

Description: The system must read source CSV data from a specified volume and load it into delta tables for customer and order data.

Preconditions:

- CSV files are available at the specified locations
(/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata and /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata)
- Delta tables for customer and order data exist or can be created

Main Flow / Functional Steps:

1. Read customer CSV data from /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata.
2. Read order CSV data from /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata.
3. Load customer data into a delta table with schema: CustId, Name, EmailId, Region.
4. Load order data into a delta table with schema: OrderId, ItemName, PricePerUnit, Qty, Date, CustId.

FR-DP-002: Clean Customer and Order Data

Title: Clean Customer and Order Data

Description: The system must remove "Null"/Null and duplicate records from both customer and order tables.

Preconditions:

- Customer and order delta tables exist and contain data

Main Flow / Functional Steps:

1. Remove rows with "Null" or Null values from the customer delta table.
2. Remove duplicate records from the customer delta table.
3. Remove rows with "Null" or Null values from the order delta table.
4. Remove duplicate records from the order delta table.

FR-DP-003: Create ordersummary Table and Load Joined Customer and Order Data

Title: Create ordersummary Table and Load Joined Customer and Order Data

Description: The system must create an "ordersummary" table if it does not exist and load joined customer and order data into it using SCD type 2.

Preconditions:

- Customer and order delta tables exist and contain clean data
- "ordersummary" table does not exist or is correctly structured

Main Flow / Functional Steps:

1. Check if the "ordersummary" table exists in catalog="gen_ai_poc_databrickscoe" and schema="sdlc_wizard".
2. If it does not exist, create the "ordersummary" table with schema: CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date.
3. Join customer and order data using the "CustId" field.
4. Load the joined data into the "ordersummary" table as an SCD type 2 table.

FR-DP-004: Update ordersummary SCD Type 2 Table on Customer Data Change

Title: Update ordersummary SCD Type 2 Table on Customer Data Change

Description: The system must update the "ordersummary" SCD type 2 table whenever there is a change in the customer table.

Preconditions:

- Customer data has changed.
- "ordersummary" table exists and is correctly populated.

Main Flow / Functional Steps:

1. Detect changes in the customer table.
2. For changed customer records, update the corresponding records in the "ordersummary" table by making old records Inactive and new records Active.
3. Update StartDate and EndDate accordingly for the affected records in "ordersummary".

FR-DP-005: Create customeraggregatespend Table and Load Aggregated Spend Data

Title: Create customeraggregatespend Table and Load Aggregated Spend Data

Description: The system must create a "customeraggregatespend" table and load aggregated spend data from the "ordersummary" table.

Preconditions:

- "ordersummary" table exists and contains data.
- "customeraggregatespend" table does not exist or is correctly structured.

Main Flow / Functional Steps:

1. Check if the "customeraggregatespend" table exists in catalog="gen_ai_poc_databrickscoe" and schema="sdhc_wizard".
2. If it does not exist, create the "customeraggregatespend" table with columns "Name", "TotalAmount", and "Date".
3. Aggregate "TotalAmount" from the "ordersummary" table, grouping by "Name" and "Date".
4. Load the aggregated data into the "customeraggregatespend" table.